Fiscal Year:	FY 2026	**Task Last Updated:**	FY 06/25/2026
PI Name:	Holian, Andrij Ph.D.		
Project Title:	Contribution of Aging to Particle-Induced Chronic Inflammation		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	59812	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2024 Space Biology NNH24ZDA001N-SBR: E.9 Space Biology Research Studies
Start Date:	04/01/2026	**End Date:**	03/31/2029
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA HQ
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Migliaccio, Christopher Ph.D. (University of Montana)		
Grant/Contract No.:	80NSSC26K0715		
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Task Description:	NASA is preparing for future manned exploration of the moon, Mars and near-Earth asteroids while minimizing any potential health effects from possible exposure to respirable-sized dusts that could be carried into living environments on space suits. Previous research used lunar and Martian regolith simulants demonstrating their toxic potential and ability to cause lung inflammation that can progress to lung fibrosis. However, the earlier work did not consider the possibility that the presence of some bioactive clays, such as montmorillonite (MMT) that is present in Martian and asteroid regolith, could add an extra permutation to health risks for future astronauts. Of additional importance, there are broad impacts of studies examining particle-induced inflammation that would benefit human health. Chronic inflammatory diseases account for over half of all mortalities worldwide and are expected to increase steadily for the next 30 years. Complicating our understand of particle-induced inflammation is the contribution of aging which is associated with altered immune responses, increased inflammation, as well as decreased immune regulation. Declining cellular function is likely to increase disease endpoints related to ongoing environmental exposures, such as space dust. Within the lung, alveolar macrophages are the first line of defense against inhaled particles for phagocytosis and clearance from the air spaces. Depending on surface characteristics phagocytosed particles can induce lysosome membrane permeabilization which is the key regulatory step leading to initiation of inflammation that is expected to change with age. In Aim 1 we will determine the relative potency, ability to initiate inflammation, and the mechanisms of toxicity of a comprehensive list of regolith simulants in vitro. In Aim 2 we will establish that regolith simulants initiate inflammation and fibrosis in young and aging male C57Bl/6 mice. We will expose C57Bl/6 mice by oropharyngeal aspiration, a well-established model of particle exposure. Studies will include conducting a time course (1 day, 7 days, 28 days, and 90 days) of inflammation and lung injury. The proposed studies will provide new information on the potential of regolith simulants to initiate inflammation that can lead to lung inflammation, fibrosis, and systemic diseases.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research promotes the highest safety standards for those who will be exploring space by ensuring that the dusts they are exposed to when removing space suits after exploratory ventures is not causing undue respiratory inflammation. We expect to define a mechanism that defines space dust induced inflammation and provides therapeutic insight into how to prevent lung inflammation in space explorers. This project will develop multiple publications, and all data collected in vitro and in vivo will be made available on the host website and NASA data repository to inform future NASA scientists.
Task Progress:	New Project for FY2026
Bibliography Type:	Description: (Last Updated:)